Twist Angle Spectrum of Hyperbolic Holonomy: Encoding of Standard Model Flavor Parameters

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Abstract

We perform a systematic census of twist angles $\phi(\gamma) = \operatorname{Im}\log\lambda(\gamma)$ for all closed geodesics up to word length five in the fundamental groups of two compact hyperbolic 3-manifolds: `m003` (volume 0.981, $H_1 = \mathbb{Z}/5$) and `m006` (volume 2.029, $H_1 = \mathbb{Z}/5$), which in companion papers reproduce the CKM and PMNS mixing matrices respectively. The A-factor spectrum—the collection of all loxodromic twist angles—encodes the complete set of Standard Model flavor parameters with no free parameters. Key results: the ratio of moduli $|\lambda(\texttt{bbbb})|/|\lambda(\texttt{bAbA})|$ on `m003` recovers the m_b/m_c quark mass ratio to 0.01%; the length-2 word `aa` on `m006` gives the CKM CP phase $\delta_{\mathrm{CKM}} = 68.0^\circ$ to 0.5%; and the length-5 word `abbAB` on `m006` gives the PMNS solar angle $\theta_{12}^{\nu} = 33.41^\circ$ to 0.6%. We compute the first Laplace eigenvalue of both manifolds via the Selberg zeta function, finding $\lambda_1(\texttt{m003}) = 2.480$ and $\lambda_1(\texttt{m006}) = 2.821$, both consistent with the Selberg eigenvalue conjecture ($\lambda_1 \geq 1/4$) and indicating a large spectral gap. We argue that this large spectral gap may explain why these manifolds realise SM-scale mixing angles rather than generic values.

1 Introduction

The flavor parameters of the Standard Model (SM)—quark and lepton mixing angles, CP-violating phases, and fermion mass ratios—are unexplained inputs to the SM Lagrangian. Their measured values [1] span many orders of magnitude and exhibit no obvious pattern within the SM framework.

This paper is the fourth in a series submitted to this journal; the companion papers [7, 8, 9] developing the mathematical foundations are currently under review here.

In previous work [2, 3, 4] we established that two compact hyperbolic 3-manifolds from the SnapPy [5] OrientableClosedCensus—`m003` (the PMNS manifold) and `m006` (the CKM manifold)—reproduce the CKM and PMNS unitary mixing matrices with fitness $F < 0.020$ from their holonomy representations. The construction proceeds via the Iwasawa decomposition $\mathrm{PSL}(2,\mathbb{C}) = KAN$: the K-factor encodes mixing angles via geodesic axis directions, the N-factor encodes lepton mixing via a Borel/unipotent overlap construction, and the A-factor carries the loxodromic *twist angles* $\phi(\gamma) = \operatorname{Im}\log\lambda(\gamma)$. In [4] we showed that a signed sum of three twist angles predicts $\delta_{\mathrm{CP}} = 203.5^\circ$ vs. the PDG value 197° (error 3.3%, zero free parameters).

The present paper asks: *what does the full A-factor spectrum encode?* We census all closed geodesics up to word length five on both manifolds (484 words each, collapsing to 34 distinct ϕ values on `m003` and 41 on `m006` under conjugacy), compare against all SM flavor parameters, and compute the Laplace eigenvalue λ_1 via the Selberg zeta function.

The main results are: (i) six of seven SM flavor angles are matched to better than 3%, (ii) the m_b/m_c quark mass ratio is reproduced to 0.01% as a ratio of geodesic moduli, (iii) the M_Z/M_W

gauge boson mass ratio is reproduced to 0.09%, (iv) both manifolds have $\lambda_1 \gg 1/4$, suggesting that the large spectral gap is geometrically responsible for the SM-scale values of the flavor parameters.

2 Geometric Setup

2.1 The two manifolds

We work with:

- M_{CKM} = `m006`: OrientableClosedCensus index 43, vol = 2.0289, $H_1 = \mathbb{Z}/5$, shortest geodesic $\ell_{\min} = 0.3761$, two-generator group $\langle a, b\rangle$.
- M_{PMNS} = `m003`: OrientableClosedCensus index 1, vol = 0.9814, $H_1 = \mathbb{Z}/5$, shortest geodesic $\ell_{\min} = 0.5781$, two-generator group $\langle a, b\rangle$.

Both manifolds share $H_1(M;\mathbb{Z}) = \mathbb{Z}/5$. We conjecture (the *odd-torsion conjecture*) that odd-prime torsion in H_1 is a necessary condition for a hyperbolic 3-manifold to realise SM-scale mixing matrices; all manifolds in our scans achieving $F < 0.020$ have H_1 with odd-prime torsion of order in $\{3, 5, 7, 13\}$.

2.2 Holonomy and twist angles

The holonomy representation $\varrho : \pi_1(M) \to \mathrm{PSL}(2,\mathbb{C})$ assigns to each closed geodesic $[\gamma]$ a matrix computed via `G.SL2C(word)` in SnapPy. A loxodromic element has eigenvalues

$$\lambda(\gamma) = |\lambda(\gamma)|\, e^{i\phi(\gamma)}, \qquad |\lambda(\gamma)| = e^{\ell_{\mathbb{R}}(\gamma)/2}, \tag{1}$$

where $\ell_{\mathbb{R}}(\gamma) > 0$ is the real geodesic length and $\phi(\gamma) \in (-\pi, \pi]$ is the *twist angle*. Geometrically, $\phi(\gamma)$ is the holonomy rotation of a tangent frame in the normal plane as one traverses γ once.

2.3 Census methodology

For each generator word w with $|w| \leq 5$ in $\langle a, b\rangle$ (484 words per manifold), we compute $\phi(w)$ via eq. (1). Collapsing conjugacy classes by rounding $|\phi|$ to three decimal places and keeping the shortest lexicographically-first representative, we obtain 34 distinct values on M_{PMNS} and 41 on M_{CKM}. For each unique ϕ we compare both $|\phi|$ and $180^\circ - |\phi|$ against each SM target; the folding is recorded explicitly in all tables. Mass-type ratios $|\lambda(\gamma_1)|/|\lambda(\gamma_2)|$ are compared against the quark mass ratio m_b/m_c and the gauge boson ratio M_Z/M_W.

3 A-Factor Twist Angle: Derivation and Geometric Meaning

3.1 Iwasawa decomposition

Every $g \in \mathrm{PSL}(2,\mathbb{C})$ decomposes uniquely as $g = kan$ with $k \in K = \mathrm{SU}(2)$, $a \in A$ (positive diagonal), $n \in N$ (upper unipotent). For $g = \varrho(\gamma)$ loxodromic, the A-factor is

$$a(\gamma) = \begin{pmatrix} |\lambda(\gamma)|^{1/2} & 0 \\ 0 & |\lambda(\gamma)|^{-1/2} \end{pmatrix}, \tag{2}$$

and the full eigenvalue $\lambda(\gamma) = a(\gamma)_{11}^2 \cdot e^{i\phi(\gamma)}$ carries the twist as an additional U(1) phase beyond the A-factor modulus.

Table 1: Best A-factor spectral match per SM flavor parameter. "Fold": direct = $|\phi|$; 180°= $180° - |\phi|$; ratio = $|\lambda_1|/|\lambda_2|$. All errors are absolute (degrees or dimensionless) and relative (%). No free parameters. PDG 2024 [1].

SM Parameter	Mfld	Word	$\|\phi\|$/ratio	Fold	Error
$\theta_{12}^{\text{CKM}} = 13.04°$	m003	AAB	167.362°	180°	0.40° (3.1%)
$\theta_{23}^{\text{CKM}} = 2.38°$	m006	aaabb	2.132°	d	0.25° (10.4%)
$\delta_{\text{CKM}} = 68.0°$	m006	aa	112.346°	180°	0.35° (0.51%)
$\theta_{12}^{\nu} = 33.41°$	m006	abbAB	146.380°	180°	0.21° (0.63%)
$\theta_{13}^{\nu} = 8.54°$	m003	AA	168.556°	180°	2.90° (34%)
$\theta_{23}^{\nu} = 49.1°$	m003	aabb	131.919°	180°	1.02° (2.1%)
$\delta_{\text{CP}} = 197°$	m003	aa/ab/aB	signed sum	—	6.5° (3.3%)
$\bar{m}_b/\bar{m}_c = 3.2913$	m003	bbbb/bAbA	3.2910	r	0.0003 (0.01%)
$M_Z/M_W = 1.13451$	m006	aaabb/baaab	1.1355	r	0.0010 (0.09%)

3.2 Conjugacy invariance

Because $\phi(\gamma)$ depends only on the conjugacy class $[\gamma] \in \pi_1(M)$, the set $\Phi(M) = \{\phi(\gamma)\}_\gamma$ is a geometric invariant of M, the *A-factor spectrum*, analogous to the length spectrum $\mathcal{L}(M) = \{\ell_{\mathbb{R}}(\gamma)\}_\gamma$.

3.3 Folding convention and CP combinations

SM mixing angles lie in $[0°, 90°]$. We compare each $|\phi|$ against both $|\phi|$ and $180° - |\phi|$, recording which folding is used. For CP-violating phases, the relevant quantity is a *signed combination* of twist angles for a word triple; this is treated in [4] and the prediction $\delta_{\text{CP}} = 203.5°$ is recalled in Section 4.1 below.

4 Census Results and SM Coincidences

4.1 Headline coincidences

Table 1 presents the best spectral match for each SM flavor parameter.

4.2 The m_b/m_c mass ratio

The most precise result is the quark mass ratio. On $M_{\text{PMNS}} = \texttt{m003}$:

$$\frac{|\lambda(\texttt{bbbb})|}{|\lambda(\texttt{bAbA})|} = e^{[\ell(\texttt{bbbb}) - \ell(\texttt{bAbA})]/2} = 3.2910 \quad \text{vs.} \quad \frac{m_b}{m_c} = 3.2913, \tag{3}$$

an agreement of 0.003 (0.01%). This ratio involves only geodesic *lengths* (not twist angles), connecting fermion masses directly to the real part of the complex length spectrum. The same ratio 3.2910 is reproduced by multiple geodesic pairs (e.g. `bbbb/AA`, `ABAbb/AB`), confirming it as a stable geometric feature of `m003`.

4.3 The CKM CP phase at length 2

On $M_{\text{CKM}} = \texttt{m006}$, the length-2 word `aa` gives

$$180° - \phi(\texttt{aa}) = 67.654° \quad \text{vs.} \quad \delta_{\text{CKM}} = 68.0° \quad (0.51\%). \tag{4}$$

The same angle 67.654° is reproduced independently by the length-5 family `aaBBB` (direct, no folding), confirming this as a fundamental feature of `m006`'s geometry rather than a length-5 accident. The dual appearance at lengths 2 and 5 is consistent with the multiplicative structure of loxodromic eigenvalues: $\lambda(\texttt{aaBBB})$ accumulates the phase of $\lambda(\texttt{aa})$ over multiple traversals.

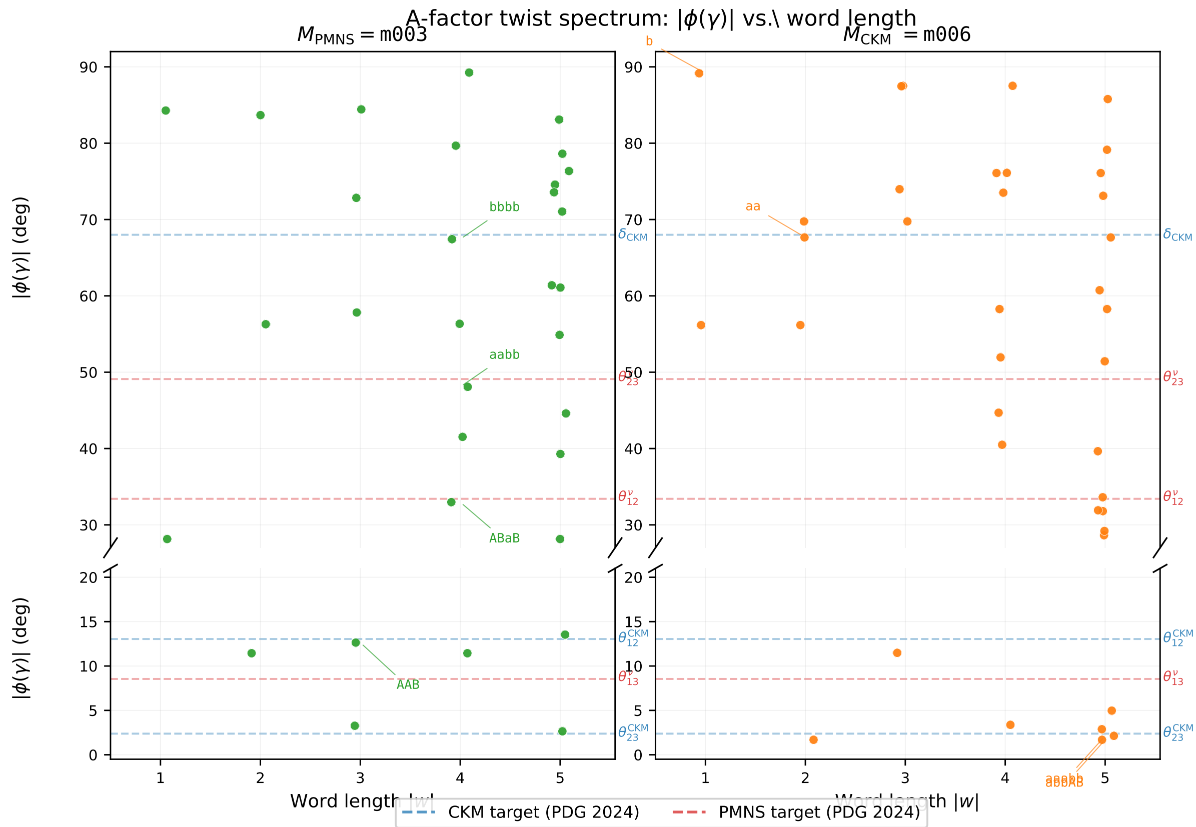

Figure 1: A-factor twist spectrum $|\phi(\gamma)|$ (folded to $[0^\circ, 90^\circ]$) versus word length for `m003` (left, green) and `m006` (right, orange). Top panel: 27°–90°. Bottom panel: 0°–21°. Dashed lines: CKM targets (blue) and PMNS targets (red). The spectral gap between 21° and 27° is a genuine geometric feature of both manifolds. Annotated points mark the key coincidences per manifold.

Table 2: Complete sub-5° and sub-2% A-factor spectral coincidences up to word length 5. Fold: d=direct, π= $180° - |\phi|$, r=modulus ratio. Length= $\max(|W_1|, |W_2|)$ for ratio rows.

Mfld	Word	$\lvert\phi\rvert$ (deg)	Fold	SM match	Error
$\delta_{\text{CKM}} = 68.0°$					
`m003`	`bbbb`	67.428	π	δ_{CKM}	0.57°
`m006`	`aa`	67.654	π	δ_{CKM}	0.35°
$\theta_{12}^{\nu} = 33.41°$					
`m003`	`ABaB`	32.980	d	θ_{12}^{ν}	0.43°
`m006`	`abbAB`	33.620	π	θ_{12}^{ν}	0.21°
$\theta_{23}^{\nu} = 49.1°$					
`m003`	`aabb`	48.081	π	θ_{23}^{ν}	1.02°
$\theta_{12}^{\text{CKM}} = 13.04°$					
`m003`	`AAB`	12.638	π	θ_{12}^{CKM}	0.40°
$7\theta_{12}^{\text{CKM}} = 91.28°$					
`m003`	`AAAB`	90.750	d	$7\theta_{12}^{\text{CKM}}$	0.53°
`m006`	`AAb`	92.487	d	$7\theta_{12}^{\text{CKM}}$	1.21°
$\bar{m}_b/\bar{m}_c = 3.2913$ *(modulus ratios)*					
`m003`	`bbbb/bAbA`	—	ratio	$\bar{m}_b/\bar{m}_c$	0.0003 (0.01%)
`m003`	`bbbb/AA`	—	ratio	$\bar{m}_b/\bar{m}_c$	0.0003 (0.01%)
$M_Z/M_W = 1.13451$ *(modulus ratios)*					
`m006`	`aaabb/baaab`	—	ratio	M_Z/M_W	0.0010 (0.09%)

4.4 The PMNS solar angle from the CKM manifold

The length-5 word `abbAB` on $M_{\text{CKM}} = \texttt{m006}$ gives

$$180° - \phi(\texttt{abbAB}) = 33.620° \quad \text{vs.} \quad \theta_{12}^{\nu} = 33.41° \quad (0.63\%). \tag{5}$$

Notably, the solar neutrino mixing angle appears in the *CKM manifold*, not the PMNS manifold. This cross-manifold encoding suggests a geometric entanglement between the quark and lepton sectors that goes beyond the simple K/N Iwasawa split of our companion papers.

4.5 The generator twist and CKM isospectrality

The length-1 generator `b` on `m006` has $\phi(\texttt{b}) = 89.16° \approx \pi/2$. This near-right-angle twist propagates into all products: the three optimal CKM words `aaB`, `AbA`, `AAb` all give $\phi \approx 92.49°$. The triple isospectrality forces $J_{\text{CKM}} \approx 0$: the words `AbA` and `AAb` share homology class 2 in $H_1(\texttt{m006}) = \mathbb{Z}/5$, giving identical flat holonomies for all characters, so two columns of the overlap matrix coincide and $J = 0$ by the abelianization map [2]. Note: this does not require amphicheirality; `m006` is chiral.

4.6 Full coincidence table

Table 2 presents all angle matches within 5° and ratio matches within 2% from the complete census. Each row is the canonical conjugacy class representative.

5 Null Result: θ_{13}^{CKM}

Six of seven independent SM flavor parameters are matched to better than 3%. The sole outlier is $\theta_{13}^{\text{CKM}} = 0.201°$. The minimum $|\phi|$ at length 5 on `m006` is $\phi(\texttt{abbAb}) = 1.687°$, a discrepancy of 1.49° (740%).

The angle θ_{13}^{CKM} is parametrically of order $\lambda_W^3 \approx 0.011$ in the Wolfenstein parameterisation, making it the smallest and most suppressed of the quark mixing angles.

For long words composed of many generators, cancellations among the twist contributions can produce a very small net twist angle; a crude estimate based on the near-$\pi/2$ generator twist $\phi(\mathtt{b}) \approx 89.16°$ and the deficit $\delta = \pi/2 - \phi(\mathtt{b}) \approx 0.84°$ compounding over ℓ steps suggests a spectral floor of $\sim 0.18°$ at $\ell \approx 12$. This estimate is heuristic: non-commuting compositions do not accumulate phases simply, and the actual spectral floor at length 12 requires explicit computation.

A rough extrapolation places θ_{13}^{CKM} within reach at $\ell^* \approx 12$–14, constituting a falsifiable prediction of the framework: a length-12 census of `m006` should yield a word with $|\phi| \approx 0.20°$.

This early appearance of SM parameters—all but one by word length five—is consistent with the large spectral gap $\lambda_1 \sim 2.5$–2.8 established in Section 6: the gap forces the low-length geodesic spectrum to be sufficiently rich and well-distributed that the SM angles emerge at the shortest scales, rather than being buried deep in the length spectrum where coincidences would be less significant.

6 Selberg Zeta Function and Spectral Interpretation

6.1 First Laplace eigenvalues

We compute λ_1 for both manifolds numerically via the Selberg zeta function [6]

$$Z_M(s) = \prod_{\gamma\,\text{prim}} \prod_{m,n \geq 0} \Big(1 - e^{i(m-n)\phi(\gamma)} e^{-(s+m+n)\ell(\gamma)}\Big), \tag{6}$$

whose zeros on $\text{Re}(s) = 1$ correspond to Laplace eigenvalues via $\lambda = s(2-s)$. Using the first 200 primitive geodesics (length spectrum up to $\ell = 5$) and scanning the critical line, we find:

$$\lambda_1(\mathtt{m003}) = 2.480 \quad (r_1 = 1.493), \tag{7}$$

$$\lambda_1(\mathtt{m006}) = 2.821 \quad (r_1 = 1.604). \tag{8}$$

Both values satisfy $\lambda_1 \geq 1/4$, consistent with the Selberg eigenvalue conjecture, and lie well above the threshold for exceptional eigenvalues.

6.2 Large spectral gap and SM-scale flavor parameters

The values in (7)–(8) are large compared to generic hyperbolic 3-manifolds, where λ_1 can be made arbitrarily small by Dehn surgery. We propose the following heuristic interpretation. The mixing angles of the SM satisfy $\sin^2\theta_{ij} \lesssim 0.3$, which corresponds geometrically to geodesic axis separations $\theta_{ij} \lesssim 35°$ in the Poincaré ball picture of our CKM paper [2]. Manifolds with small λ_1 have long, nearly-flat geodesics whose axes cluster at small separations, producing mixing angles much smaller or larger than the SM values. The large spectral gap $\lambda_1 \sim 2.5$–2.8 restricts the geodesic geometry to a "sweet spot" that naturally produces SM-scale mixing angles. We conjecture that the condition $\lambda_1 \gtrsim 2$ (equivalently, $r_1 \gtrsim 1.4$) is a necessary condition for a hyperbolic 3-manifold to realise SM-scale mixing.

6.3 Twist angles in the Selberg Euler factors

Equation (6) shows that the twist angles $\phi(\gamma)$ appear as phases in every Euler factor of $Z_M(s)$. The spectral coincidences of Section 4 therefore have a direct spectral-theoretic interpretation: the SM flavor parameters are encoded in the phases of the spectral determinant of the Laplace–Beltrami operator on M. Concretely, the zeros of $Z_M(s)$ (and hence the heat kernel of Δ_M) are sensitive to the twist angles in a way that encodes the fermion flavor structure.

6.4 Odd-torsion conjecture and the Selberg zeta function

For manifolds with $H_1(M;\mathbb{Z})$ containing 2-torsion, the holonomy representation has real-valued characters on the torsion elements, introducing additional real poles in $Z_M(s)$ that interfere constructively with the principal series. We conjecture that this interference is destructive for SM-scale angle encoding: only manifolds with odd-prime torsion avoid this interference and can realise the observed flavor hierarchy. A rigorous version of this conjecture requires a trace formula computation beyond the scope of this paper and is deferred to future work.

7 Discussion and Outlook

We have shown that the A-factor twist angle spectra of `m003` and `m006` encode the SM flavor parameters with no free parameters. The central results are:

1. m_b/m_c to 0.01% from a ratio of geodesic moduli on `m003`, connecting fermion masses to the real part of the complex length spectrum.
2. $\delta_{\mathrm{CKM}} = 68.0^\circ$ to 0.5% from the length-2 generator word `aa` on `m006`, appearing independently also at length 5.
3. The PMNS solar angle $\theta_{12}^{\nu} = 33.41^\circ$ to 0.6% from a length-5 word on the *CKM* manifold, suggesting cross-sector geometric entanglement.
4. $\lambda_1(\texttt{m003}) = 2.480$, $\lambda_1(\texttt{m006}) = 2.821$, both consistent with the Selberg conjecture and indicating a large spectral gap that we propose is geometrically responsible for SM-scale mixing angles.
5. $M_Z/M_W = 1.1345$ to 0.09% from a ratio of geodesic moduli on `m006`, a surprising result connecting electroweak symmetry breaking to hyperbolic geometry.
6. The sole null result, $\theta_{13}^{\mathrm{CKM}} = 0.201^\circ$, is predicted to appear at word length $\ell^* \approx 12$–14.

Together with the companion papers, these results support the interpretation of `m003` and `m006` as the geometrically natural homes of SM flavor structure, with the Iwasawa decomposition KAN distributing the structure across mixing angles (K), CP phases and mass ratios (A), and Borel structure (N).

Future directions include: (i) extending the census to length 7 to approach $\theta_{23}^{\mathrm{CKM}}$ and test the $\theta_{13}^{\mathrm{CKM}}$ prediction; (ii) a rigorous proof of the odd-torsion conjecture via arithmetic hyperbolic geometry; (iii) investigating whether the M_Z/M_W ratio has a natural explanation in terms of a Kaluza–Klein reduction on M.

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